



SUJET STATAPP 364 - 2026-2027

Sujet proposé par : Belpaire Michael - michael.belpaire@symbioticsgroup.com

Organisme : Symbiotics Asset Management

Catégorie : Finance

Sujet :

Statistical Modelling, Stress Testing and Optimal Structuring of a Blended Finance Fund

Description du sujet :

Blended finance uses concessional or catalytic capital to mobilise private capital towards projects with development, climate or social objectives. A common structure consists of a junior, first-loss tranche funded by a public or philanthropic sponsor, and a senior tranche offered to private investors. The junior tranche absorbs losses first, thereby improving the risk-return profile of the senior tranche.

The central structuring challenge is to determine the appropriate size and design of each tranche. A large junior tranche offers strong protection to senior investors but reduces the mobilisation of private capital; a thin junior tranche increases leverage but can expose the senior tranche to excessive losses or cash-flow shortfalls.

This project will develop a transparent statistical framework for a stylised closed-end blended finance fund. The fund will invest in a diversified portfolio of impact-oriented debt instruments, loans or projects. Following the framework of Ben Slimane et al. (2025), the asset portfolio will be treated as largely exogenous, reflecting a predefined impact mandate, while the liability-side capital structure will be calibrated to the portfolio's risk characteristics.

The main research question is:

Given a predefined portfolio of impact assets, what junior-senior capital structure maximises private-capital mobilisation while satisfying senior-investor risk, cash-flow and credit-quality constraints?

The project will focus on four connected tasks:

? Modelling portfolio credit risk through default probabilities, recovery rates and dependence between assets;

? Simulating fund and tranche cash flows under a contractual payment waterfall;

? Quantifying the sensitivity of tranche outcomes to key risk drivers and adverse stress scenarios;

? Optimising the capital stack under explicit financial and impact-related constraints.

The core model will use a two-tranche structure: junior/first-loss and senior. One extension may then be selected according to the group's progress:

? introducing a mezzanine tranche;

? testing a senior-protection mechanism, such as a reserve account or loss carryforward;

? incorporating a simplified currency or interest-rate stress.

This restriction is intentional: it enables students to build, validate and interpret one robust model rather than producing an overly broad catalogue of methods

Données du sujet :

To be defined

Modèles envisagés du sujet :

to be defined

Bibliographie du sujet :

Ben Slimane, M., Dumas, J.-M., Lekhel, A. and Roncalli, T. (2025), A Framework for Structuring a Blended Finance Fund, Amundi Investment Institute, Working Paper 175.

OECD (2025), OECD DAC Blended Finance Guidance 2025.

Glasserman, P. (2003), Monte Carlo Methods in Financial Engineering.

McNeil, A., Frey, R. and Embrechts, P. (2015), Quantitative Risk Management.

Publicly available structured-finance rating criteria and methodology notes from Fitch Ratings, Moody's Ratings and S&P Global Ratings..

Notation du sujet :

Commentaires :